

tempssm Reference Manual

Version 0.3.0

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Summary

tempssm provides a practical R interface for state-space analysis of temperature time series. The package focuses on linear Gaussian state-space models estimated by Kalman filtering and smoothing, using the **KFAS** package as the computational backend (Helske, 2017).

Key Features

- Fits linear Gaussian state-space models to temperature time series.
- Represents temperature dynamics using interpretable latent components: long-term trend, seasonal variation, autoregressive structure, and optional exogenous effects.
- Supports arbitrary seasonal frequencies, while the current examples and validation focus primarily on monthly temperature data.
- Allows the autoregressive order to be specified by the user.
- Provides S3 methods for summaries, diagnostics, and plots.
- Includes time-series cross-validation tools for model evaluation.

Prior Art and Scope

tempssm provides a domain-focused workflow for analyzing temperature time series with linear Gaussian state-space models. It brings together model construction, component extraction, uncertainty summaries, residual diagnostics, visualization, and time-series cross-validation in a single R package interface tailored to temperature applications.

The package builds on established statistical methodology, including linear Gaussian state-space modeling, Kalman filtering, and Kalman smoothing. Model estimation is handled through the **KFAS** package, which provides a general framework for state-space models in R.

The initial implementation was adapted from the supplementary code provided by Baba (2024), accompanying Baba et al. (2024), which analyzed sea temperature trends using a linear Gaussian state-space model. The supplementary code is publicly available at:

<https://github.com/logics-of-blue/sea-temperature-trend-jogashima>

Compared with that prior implementation, **tempssm** extends the workflow into a reusable R package interface with input validation, documented S3 methods, tests, diagnostics, cross-validation utilities, and examples for broader temperature time-series analysis.

The next section describes the state-space model used by **tempssm** and clarifies how the trend, seasonal, autoregressive, and exogenous components are represented.

State-Space Model in tempssm

We consider an extended version of the Basic Structural Time Series Model (BSTSM) to describe temperature time series with explicit long-term trend, seasonal variability, and auto-regressive component. The observation equation is given by

$$y_t = \alpha_t + v_t. \quad (1)$$

where t denotes the time index, y_t is the observed temperature, α_t is the latent state, and v_t is the observation error.

The latent state is decomposed as

$$\alpha_t = \mu_t + s_t + r_t + E_t. \quad (2)$$

where μ_t is the long-term trend component, s_t is the seasonal component, r_t is a stationary autoregressive component, and E_t is the contribution of exogenous variables.

The level component follows a second-order stochastic process:

$$\mu_t = 2\mu_{t-1} - \mu_{t-2} + \zeta_t, \quad \zeta_t \sim \mathcal{N}(0, \sigma_\zeta^2). \quad (3)$$

where ζ_t is the process error of the long-term trend component.

The seasonal component is modeled with a sum-to-zero constraint:

$$s_t = - \sum_{i=t-f}^{t-1} s_i + \omega_t, \quad \omega_t \sim \mathcal{N}(0, \sigma_\omega^2). \quad (4)$$

where ω_t is the process error of the seasonal component and f denotes the seasonal frequency (for example, $f = 12$ for monthly data). This formulation makes seasonal effects identifiable and enables models to be constructed for time series with different temporal resolutions. By imposing the sum-to-zero constraint over one complete seasonal cycle, the seasonal component captures recurring deviations from the underlying long-term trend without introducing long-term drift. In models without a seasonal component, the seasonal term s_t is set to zero and omitted from the state equation.

The autoregressive component follows an l th-order autoregressive (AR) process:

$$r_t = \phi_1 r_{t-1} + \phi_2 r_{t-2} + \dots + \phi_l r_{t-l} + \tau_t, \quad \tau_t \sim \mathcal{N}(0, \sigma_\tau^2). \quad (5)$$

Here, τ_t denotes the process error of the autoregressive component. The process error terms (ζ_t , ω_t , and τ_t) are assumed to be mutually independent and normally distributed.

The exogenous component is defined as

$$E_t = \beta_1 x_{1,t} + \beta_2 x_{2,t} + \dots + \beta_m x_{m,t}. \quad (6)$$

The exogenous term E_t represents the influence of external factors. These may include large-scale climate indices, regional environmental variables, or other physically motivated predictors relevant to the observed temperature time series. Each exogenous variable enters the model linearly through a time-invariant regression coefficient ($\beta_1, \beta_2, \dots, \beta_m$), allowing the magnitude and direction of its contribution to be estimated jointly with the latent state components.

When applying the model to observational data, the total number of parameters to be estimated (k) depends on the order of the autoregressive component and the number of exogenous variables included in the model. For example, in a model with a second-order autoregressive component and no exogenous covariates, six parameters are estimated: the observation error variance, the process error variances associated with the long-term trend, seasonal, and autoregressive components, and the first- and second-order autoregressive coefficients (ϕ_1 and ϕ_2). In the general case with an l th-order autoregressive component and m exogenous variables, the total number of parameters is $k = 4 + l + m$.

The core implementation of parameter estimation procedure is based on the supplementary code provided by Baba et al. (2024). Parameter estimation is performed using a two-step optimization strategy recommended

by Helske (2017). In the first step, model parameters are estimated using the Nelder-Mead method (Nelder & Mead, 1965) with user-specified initial values. The resulting estimates are then used as initial values in a second optimization step based on the BFGS algorithm (Shanno, 1970). The main model-fitting function, `tempssm()`, returns both filtering and smoothing estimates. Unless otherwise stated, all results presented in this vignette are based on the smoothed estimates.

How to Use

Set Environment

Load the following R packages to run the examples below. If any packages are not installed, please install them as needed.

```
## Set libraries
library(tempssm)
library(forecast)

library(purrr)
library(tibble)
library(readr)
library(dplyr)
library(ggplot2)

library(patchwork)
```

Input Data Format

Input data for `tempssm` must be supplied as an R `ts` object, which represents a regularly spaced time series (see `?stats::ts` or <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/ts.html>).

To support data preparation, the package includes utility functions that convert external observational data into `ts` objects (see Appendix).

Exercise I: Applying a State-Space Model to a Univariate Temperature Time Series

Objective

In Exercise I, we apply a linear Gaussian state-space model to a univariate temperature time series. This exercise introduces basic modeling without exogenous variables and examines the role of autoregressive dynamics.

Loading the Sea Surface Temperature (SST) Dataset

A sample sea surface temperature (SST) dataset is included in the package.

- **Dataset:** Monthly sea surface temperature (SST) off Yamaguchi Pref., Japan
- **Unit:** °C
- **Period:** February 2002 to December 2023

This dataset was created by aggregating daily SST data obtained from the Japan Meteorological Agency website (<https://www.jma.go.jp/jma/indexe.html>) into monthly values.

```
data(yamaguchi_sst) # load a ts object of SST off Niigata
head(yamaguchi_sst)
```

```
##           Jan      Feb      Mar      Apr      May      Jun
## 1982 14.85516 13.36500 13.55645 14.29933 17.72419 21.53000
```

```
summary(yamaguchi_sst)
```

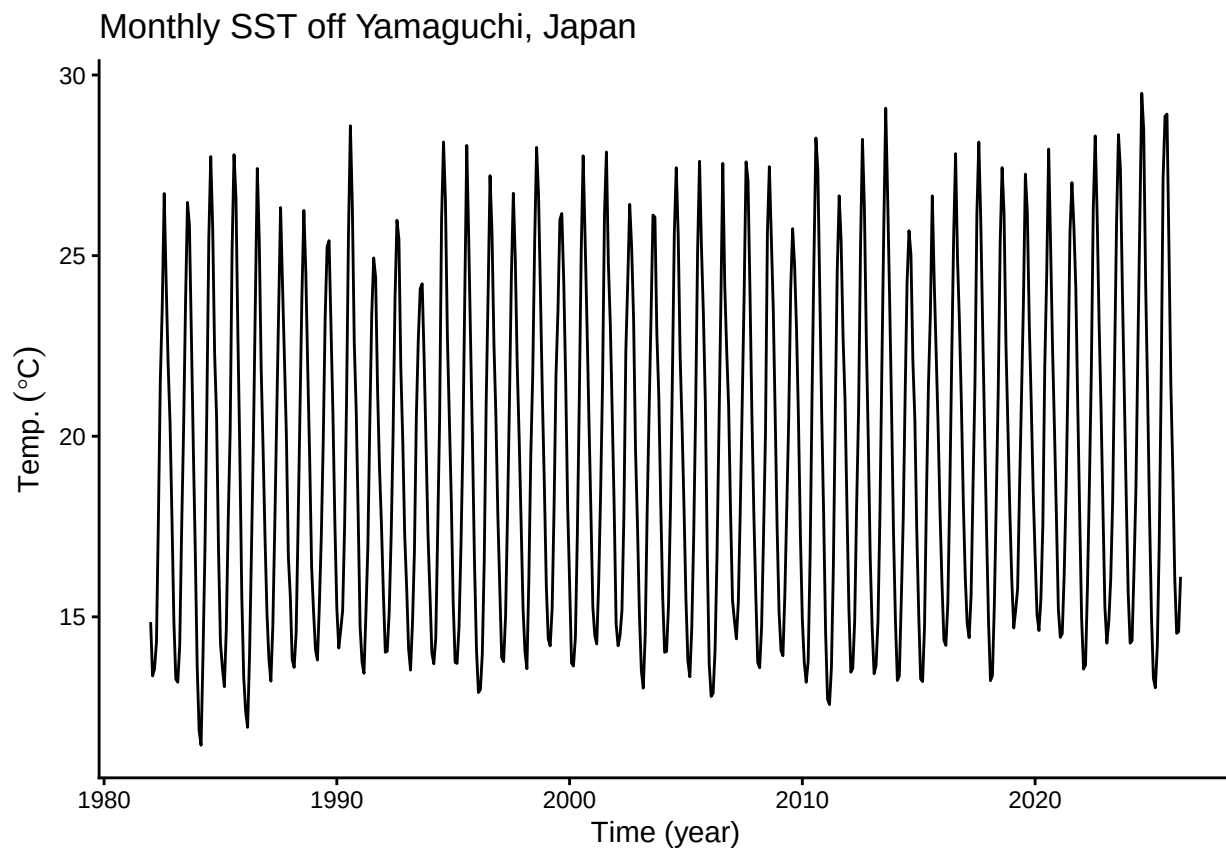
```
##      Temp
## Min.   :11.45
## 1st Qu.:15.13
## Median :18.84
## Mean   :19.54
## 3rd Qu.:23.63
## Max.   :29.49
```

Plotting the Monthly SST Time Series

We begin by visualizing the monthly SST time series to examine its overall structure, including apparent trends, seasonal variability, and the presence or absence of missing observations.

```
plt_yamaguchi_sst <- forecast::autoplot(yamaguchi_sst) +
  labs(y = expression(Temp.~(degree*C)),
       x = "Time (year)" ) +
  ggtitle("Monthly SST off Yamaguchi, Japan") +
  theme_classic()

plot(plt_yamaguchi_sst)
```



The overall mean SST is approximately 19.5 °C, and a clear seasonal pattern is visible. The series contains no missing observations. Although SST appears to increase over the study period, year-to-year variability is also evident, making the long-term trend difficult to assess from the raw time series alone.

Applying a Linear Gaussian State-Space Model

When a `ts` object containing temperature time-series data (here, `yamaguchi_sst`) is passed to the core function `tempssm()`, model construction and parameter estimation are performed together. The returned S3 object of class `tempssm` (here, `res_ar1`) stores the filtering and smoothing estimates, as well as the constructed model and input data. By default, `tempssm()` fits a first-order autoregressive model.

```
# model with first-order autoregressive component
res_ar1 <- tempssm(yamaguchi_sst) # (ar_order=1: default)
summary(res_ar1)
```

```
## tempssm summary
## -----
## Call:
## tempssm(temp_data = yamaguchi_sst)
##
## Model fit:
##   Likelihood type: marginal
##   Log-likelihood : -437.2
##   k               : 5
##   Diffuse states : 13
##   Converged      : TRUE
##
## Variance parameters:
##   Observation (H): 4.542027e-14
##   State (Q trend): 9.936473e-08
##   State (Q season): 8.408352e-56
##   State (Q ar): 0.3145626
##
## Components of auto-regression:
##   Order of AR: 1
##   Coefficient of AR1: 0.6202321
```

From the summary output, confirm that the model has converged (**Converged: TRUE**). The output also reports statistics such as the number of parameters (**k**), the log-likelihood, the likelihood type, and the number of diffuse initial states. The parameter estimates include the observation error variance (**H**), the process error variance of the long-term trend component (**Q trend**), the process error variance of the seasonal component (**Q season**), the process error variance of the autoregressive component, and the first-order autoregressive coefficient (**AR1**).

The log-likelihood and the associated number of estimated parameters can also be extracted directly from the fitted `tempssm` object using `logLik()`.

```
ll <- logLik(res_ar1)
ll

## 'log Lik.' -437.1989 (df=5)
attr(,"df") # number of parameters
```

```
## [1] 5
```

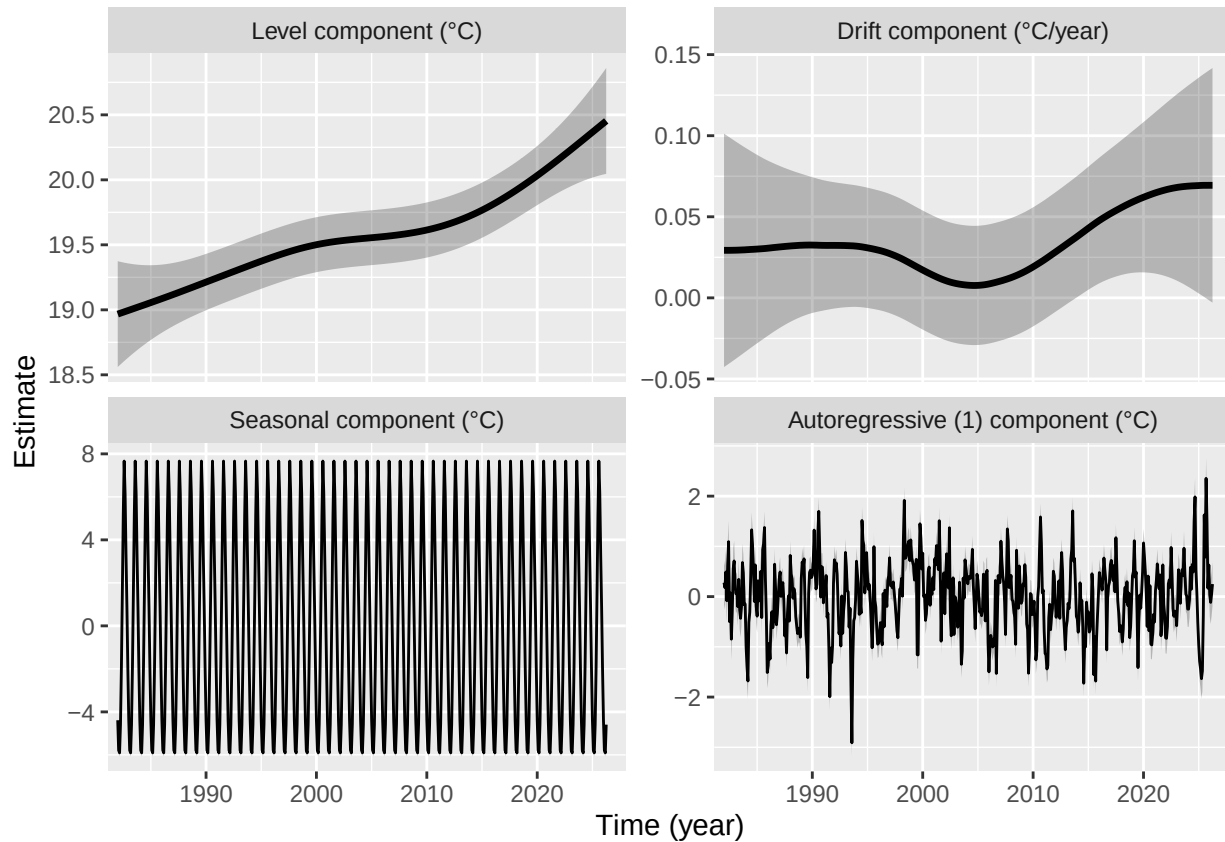
By default, `tempssm()` uses the KFAS marginal likelihood for parameter estimation. The diffuse likelihood remains available by explicitly setting `marginal = FALSE`. The selected likelihood type is stored in the fitted `tempssm` object and is used consistently by downstream methods such as `logLik()` and `summary()`.

The package intentionally does not compute AIC for `tempssm` objects. The log-likelihood and parameter count remain available through `logLik()` for users who need them for their own model-assessment workflows.

Plotting Long-Term Trend, Drift, Seasonal, and Autoregressive Components

We extract the corresponding latent components from the state-space model and visualize the estimated long-term evolution of temperature levels and their rate of change (drift). Plotting seasonal variation and autoregressive dependence as well makes the underlying trend structure easier to examine.

```
# plot all components at once  
plot(res_ar1)
```



The long-term trend in the upper-left panel indicates an increasing SST pattern over the study period. In this example, the average annual rate of SST increase over the study period is approximately 0.036 °C.

However, the rate of SST change itself also appears to vary over time. Focusing on the rate of change (drift) in the upper-right panel, the rate of change declines to zero or near zero during the 2000s and then increases after 2010. The shaded gray areas represent 95% confidence intervals for the estimated latent states and illustrate the uncertainty associated with each estimated component.

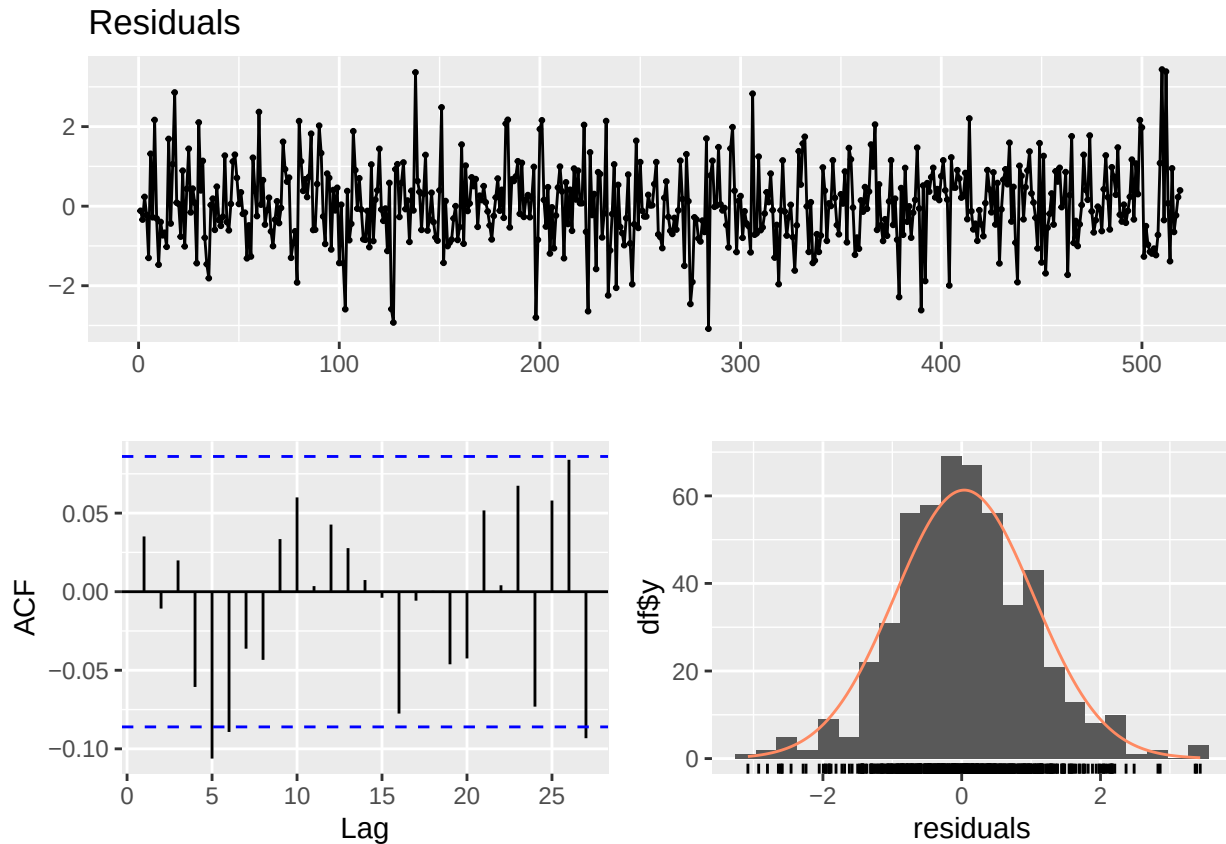
The standard plotting interface is `plot(res)`. The ggplot2-style interface `autoplot(res)` is also available and produces the same component plot by default. To obtain a ggplot object for an individual component, specify the `component` argument in `autoplot()` as follows.

```
# plot each of components at once  
plt_level <- autoplot(res_ar1, component = c("level"))  
plt_drift <- autoplot(res_ar1, component = c("drift"))  
plt_season <- autoplot(res_ar1, component = c("season"))  
plt_ar1 <- autoplot(res_ar1, component = c("ar1"))
```

Model Diagnostics

The package provides diagnostic tools for checking whether the fitted model has left notable structure in the residuals. In particular, residual time-series plots, residual autocorrelation, residual distributions, and Ljung-Box tests can be used to assess remaining temporal dependence and departures from the Gaussian error assumption.

```
plot_tempsm_residual_diagnostics(res_ar1)
```



In the model diagnostic plot, the upper panel shows the residual time series, the lower-left panel shows the residual autocorrelation plot (ACF plot), and the lower-right panel shows the residual frequency distribution. These plots should be checked for any notable residual patterns.

In this example, the ACF plot shows an isolated spike around lag 5 and lag 27, but there is no clear sequence of successive significant lags or repeated seasonal pattern. Such isolated spikes can occur when many lags are inspected, so they should be interpreted together with the Ljung-Box test and the residual time-series plot rather than used as the sole basis for changing the model.

```
diag <- diagnose_residuals(res_ar1)
print(diag)
```

```
## # A tibble: 1 x 4
##   lb_stat lb_lag lb_pvalue kurtosis
##   <dbl>   <dbl>   <dbl>   <dbl>
## 1    18.1     12     0.111     3.70
```

The `lb_stat`, `lb_lag`, and `lb_pvalue` columns returned by `diagnose_residuals()` correspond to the Ljung-Box test statistic, the lag used in the test, and the corresponding P-value. For monthly time series, the default Ljung-Box lag is the seasonal frequency, so the default test evaluates residual autocorrelation up to lag 12. In this example, the Ljung-Box test does not indicate significant residual autocorrelation up to lag 12.

If residual autocorrelation at longer lags is of concern, the lag can be specified manually. For example, the following code repeats the Ljung-Box test up to lags 24 and 36.

```
diag_lag24 <- diagnose_residuals(res_ar1, lb_lag = 24)
diag_lag36 <- diagnose_residuals(res_ar1, lb_lag = 36)

diag_lags <- rbind(diag, diag_lag24, diag_lag36)
diag_lags$check <- c("lag 12", "lag 24", "lag 36")
diag_lags[, c("check", "lb_stat", "lb_lag", "lb_pvalue", "kurtosis")]
```

```
## # A tibble: 3 x 5
##   check lb_stat lb_lag lb_pvalue kurtosis
##   <chr>   <dbl> <dbl>   <dbl>   <dbl>
## 1 lag 12  18.1    12    0.111    3.70
## 2 lag 24  30.8    24    0.158    3.70
## 3 lag 36  48.9    36    0.0746   3.70
```

These diagnostics should be interpreted together with the residual ACF plot. A small number of isolated ACF spikes can occur by chance when many lags are inspected, whereas repeated or periodic spikes may suggest remaining temporal structure.

Examining the Autoregressive (AR) Order

In this manual, the default AR(1) specification is used as the working model. The autoregressive component is included to absorb short-term serial dependence that is not represented by the latent level, drift, and seasonal components. A higher AR order should therefore not be treated as an automatic improvement. Instead, the AR order can be adjusted when residual diagnostics suggest that notable autocorrelation remains.

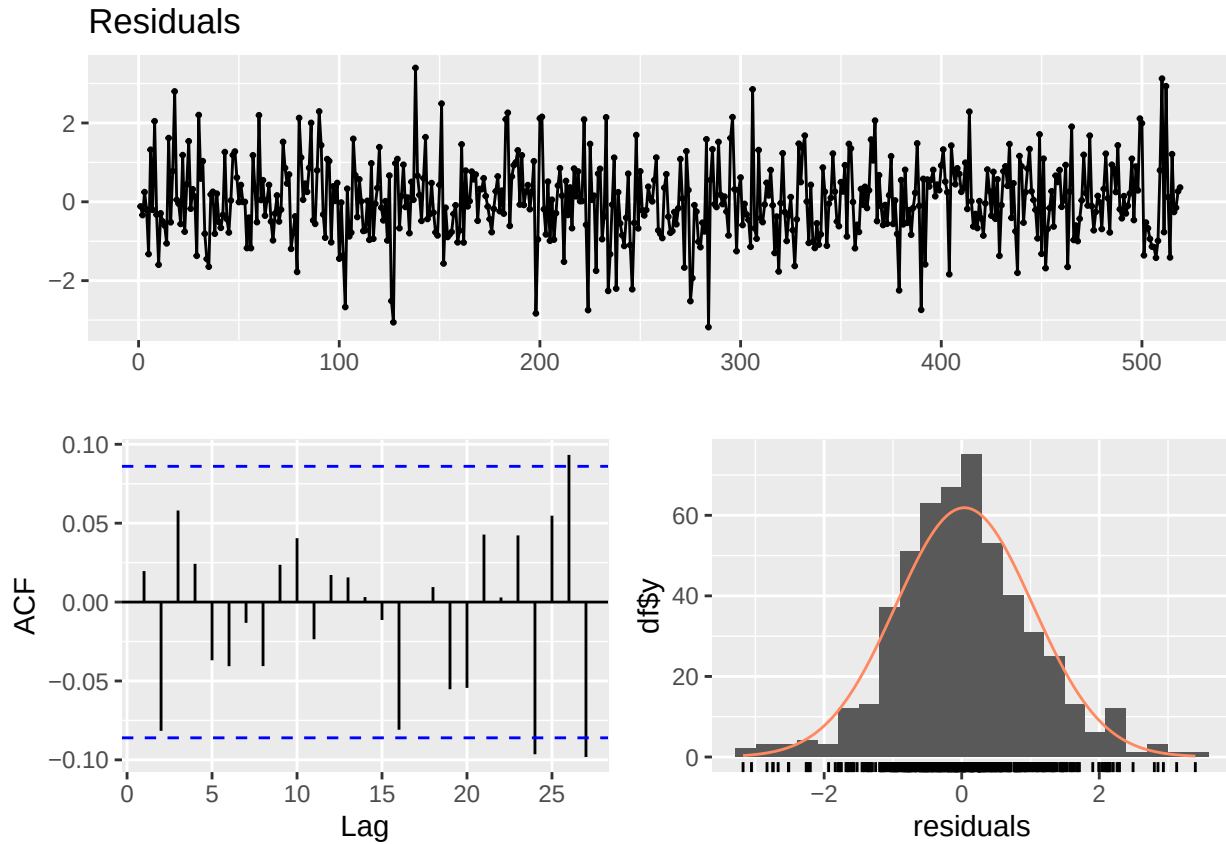
For example, if residual autocorrelation remains after fitting the AR(1) model, users can fit an AR(2) model and repeat the same diagnostics.

```
# Optional sensitivity check with a second-order autoregressive component
res_ar2 <- tempssm(yamaguchi_sst, ar_order = 2)
summary(res_ar2)
```

```
## tempssm summary
## -----
## Call:
## tempssm(temp_data = yamaguchi_sst, ar_order = 2)
##
## Model fit:
##   Likelihood type: marginal
##   Log-likelihood : -434.26
##   k               : 6
##   Diffuse states : 13
##   Converged      : TRUE
##
## Variance parameters:
##   Observation (H): 0.120908
##   State (Q trend): 2.206262e-07
##   State (Q season): 5.525557e-16
##   State (Q ar): 0.1044204
##
## Components of auto-regression:
##   Order of AR: 2
##   Coefficient of AR1: 1.185818
##   Coefficient of AR2: -0.4790522
```



```
plot_tempssm_residual_diagnostics(res_ar2)
```



```
diag_lag12_ar2 <- diagnose_residuals(res_ar2, lb_lag = 12)
diag_lag24_ar2 <- diagnose_residuals(res_ar2, lb_lag = 24)
diag_lag36_ar2 <- diagnose_residuals(res_ar2, lb_lag = 36)
diag_lags_ar2 <- rbind(diag_lag12_ar2, diag_lag24_ar2, diag_lag36_ar2)
diag_lags_ar2$check <- c("lag 12", "lag 24", "lag 36")
diag_lags_ar2[, c("check", "lb_stat", "lb_lag", "lb_pvalue", "kurtosis")]
```

```
## # A tibble: 3 x 5
##   check lb_stat lb_lag lb_pvalue kurtosis
##   <chr>   <dbl> <dbl>   <dbl>   <dbl>
## 1 lag 12    9.93    12    0.623    3.66
## 2 lag 24   24.0    24    0.461    3.66
## 3 lag 36   43.9    36    0.170    3.66
```

In a sensitivity check with an AR(2) specification, the short-lag residual autocorrelation was also limited, but the residual ACF showed a negative spike around lag 24, 26, 27. This pattern suggests that increasing the AR order does not necessarily improve all aspects of the residual structure. Therefore, the AR(2) model should be treated as a model requiring additional diagnostic attention rather than as an automatic improvement over the AR(1) baseline.

In the present example, the residual diagnostics do not provide strong evidence that a higher AR order is needed, so we proceed with the AR(1) model as the baseline specification.

Estimated Parameters and Latent-State Components

The long-term trend component and its rate of change (drift) can be extracted as `ts` objects as follows. `r`

```
# Smoothing estimates
```

```
alpha_hat <- res_ar1$kfs$alphahat
head(alpha_hat)
```

```
##           level      slope sea_dummy1 sea_dummy2 sea_dummy3 sea_dummy4
## Jan 1982 18.96708 0.002442910 -4.388514 -1.989138 0.8440593 3.3269013
## Feb 1982 18.96952 0.002442962 -5.783514 -4.388514 -1.9891376 0.8440593
## Mar 1982 18.97197 0.002442995 -5.905440 -5.783514 -4.3885143 -1.9891376
## Apr 1982 18.97441 0.002443203 -4.593786 -5.905440 -5.7835140 -4.3885143
## May 1982 18.97685 0.002443328 -1.903379 -4.593786 -5.9054404 -5.7835140
## Jun 1982 18.97930 0.002443456 1.457171 -1.903379 -4.5937863 -5.9054404
##           sea_dummy5 sea_dummy6 sea_dummy7 sea_dummy8 sea_dummy9 sea_dummy10
## Jan 1982 6.1507774 7.6637613 5.1211012 1.4571706 -1.903379 -4.593786
## Feb 1982 3.3269013 6.1507774 7.6637613 5.1211012 1.457171 -1.903379
## Mar 1982 0.8440593 3.3269013 6.1507774 7.6637613 5.121101 1.457171
## Apr 1982 -1.9891376 0.8440593 3.3269013 6.1507774 7.663761 5.121101
## May 1982 -4.3885143 -1.9891376 0.8440593 3.3269013 6.150777 7.663761
## Jun 1982 -5.7835140 -4.3885143 -1.9891376 0.8440593 3.326901 6.150777
##           sea_dummy11      arima1
## Jan 1982 -5.905440 0.27659369
## Feb 1982 -4.593786 0.17898925
## Mar 1982 -1.903379 0.48992423
## Apr 1982 1.457171 -0.08129112
## May 1982 5.121101 0.65071819
## Jun 1982 7.663761 1.09353211
```

```
# Smoothing estimate of level component
```

```
level_ts <- get_level_ts(res_ar1)
```

```
# Smoothing estimate of drift component
```

```
drift_ts <- get_drift_ts(res_ar1)
```

```
# Average drift rate per year across the full period
```

```
mean_drift_year <- mean(drift_ts)
```

```
print(mean_drift_year)
```

```
## [1] 0.03365529
```

The average annual rate of SST increase was estimated to be 0.0366 °C.

Short-Term Prediction

A fitted `tempssm` object can also be passed to `predict()` to obtain short-term predictions. By default, `predict(res)` returns a one-step-ahead prediction beyond the end of the observed series. This is useful for visual checks of how the fitted model extrapolates the estimated level, seasonal, and autoregressive components.

```
pred_1 <- predict(res_ar1)
```

```
pred_1
```

```
##           May
```

```
## 2026 18.71037
```

Predictions for multiple future time points can be requested by setting the `n.ahead` argument.

```
pred_12 <- predict(res_ar1, n.ahead = 12)
```

```
pred_12
```

```
##           Jan       Feb       Mar       Apr       May       Jun       Jul       Aug
## 2026                                18.71037 22.01798 25.65128 28.17714
## 2027 16.12028 14.72978 14.61284 15.92978
##           Sep       Oct       Nov       Dec
## 2026 26.65593 23.82915 21.34670 18.51594
## 2027
```

Prediction uncertainty can also be returned by setting the `interval` argument. With `interval = "confidence"`, the returned lower and upper bounds describe uncertainty in the predicted mean response. With `interval = "prediction"`, the bounds describe uncertainty in a future observation and therefore also include observation error; prediction intervals are usually wider than confidence intervals. The confidence level is controlled by the `level` argument and defaults to 0.95.

```
pred_12_pi <- predict(
  res_ar1,
  n.ahead = 12,
  interval = "prediction",
  level = 0.95
)
```

```
pred_12_pi
```

```
##           fit       lwr       upr
## May 2026 18.71037 17.58819 19.83255
## Jun 2026 22.01798 20.68625 23.34972
## Jul 2026 25.65128 24.24035 27.06221
## Aug 2026 28.17714 26.73244 29.62184
## Sep 2026 26.65593 25.19542 28.11643
## Oct 2026 23.82915 22.36046 25.29783
## Nov 2026 21.34670 19.87329 22.82011
## Dec 2026 18.51594 17.03948 19.99240
## Jan 2027 16.12028 14.64169 17.59886
## Feb 2027 14.72978 13.24938 16.21017
## Mar 2027 14.61284 13.13087 16.09480
## Apr 2027 15.92978 14.44637 17.41319
```

These predictions should be interpreted as model-based extrapolations rather than definitive forecasts. Uncertainty generally increases as the prediction horizon becomes longer, and long-horizon predictions can be sensitive to model assumptions about trend, seasonality, and autoregressive dependence.

Exercise II: Applying a State-Space Model to a Temperature Time Series with an Exogenous Variable

Objective

We extend the state-space modeling framework to examine the effect of an exogenous factor on temperature variation. Specifically, we examine the effect of the Pacific Decadal Oscillation (PDO) as an exogenous variable on SST observed off Yamaguchi Prefecture, Japan.

Loading the PDO Index Dataset: PDO Index as an Exogenous Variable

- **Data:** Monthly Pacific Decadal Oscillation (PDO) index (JMA)
- **Period:** January 1901 to December 2025

The Pacific Decadal Oscillation (PDO) index is defined as the projection of monthly mean sea surface temperature (SST) anomalies onto the leading empirical orthogonal function (EOF) of SST variability over

the North Pacific north of 20°N. The EOF is computed using SST anomalies for 1901–2000, defined relative to the 1901–2000 monthly climatology. To remove the global warming signal, the global-mean SST anomaly is subtracted from each grid point prior to the EOF analysis. In this package, we use the PDO index provided by the Japan Meteorological Agency (JMA), available at https://www.data.jma.go.jp/kaiyou/data/shindan/b_1/pdo/pdo.txt.

```
data(pdo) # load a ts object of PDO index
head(pdo)
```

```
##           Jan      Feb      Mar      Apr      May      Jun
## 1901  1.0040  0.7403  0.9011 -0.0109 -0.2325 -0.6810
```

Intersecting the Temperature and PDO Time Series

For state-space modeling with exogenous variables, all input time series must share a common and aligned time index. In this step, the temperature and PDO time series are restricted to their overlapping period by trimming the leading and trailing portions so that both datasets cover the same time span.

The function `tempssm::trim_ts_overlap()` is used to align the two `ts` objects on a shared timeline and returns time series containing only the common period.

```
# Generate an object on a shared timeline
trimmed_series <- trim_ts_overlap(yamaguchi_sst,
                                  pdo,
                                  temp_name = "Temp",
                                  exo_name = "PDO")

yamaguchi_sst_trim <- trimmed_series$temperature
pdo_trim <- trimmed_series$exogenous

start(yamaguchi_sst_trim)
```

```
## [1] 1982    1
end(yamaguchi_sst_trim)
```

```
## [1] 2025   12
start(pdo_trim)
```

```
## [1] 1982    1
end(pdo_trim)
```

```
## [1] 2025   12
```

Plotting Time Series of SST and the PDO Index

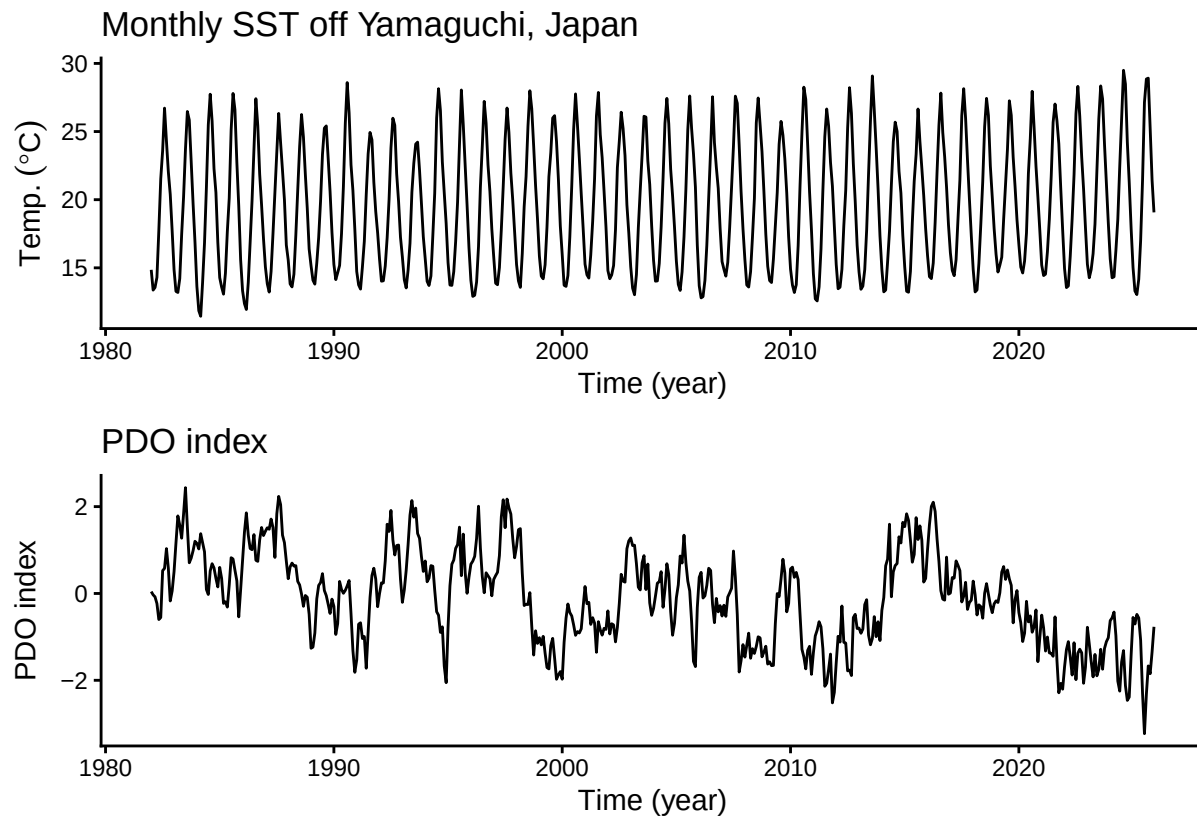
We visualize the time-series structures of SST and the PDO index.

```
plt_yamaguchi_sst_trim <- forecast::autoplot(yamaguchi_sst_trim) +
  labs(y = expression(Temp.~(degree*C)),
       x = "Time (year)") +
  ggtitle("Monthly SST off Yamaguchi, Japan") +
  theme_classic()

plt_pdo <- forecast::autoplot(pdo_trim) +
  labs(x = "Time (year)", y = "PDO index") +
```

```
ggtitle("PDO index") +
theme_classic()

plt_yamaguchi_sst_trim + plt_pdo + patchwork::plot_layout(ncol=1)
```



The PDO index data used in this exercise contain no missing values. When using your own exogenous-variable data, however, make sure that the exogenous variables do not contain missing values before fitting the model. `tempssm()` does not allow missing values in exogenous variables, and model fitting will stop with an error if such data are supplied. If necessary, apply appropriate preprocessing, such as imputation, before the analysis.

In contrast, missing values in the dependent temperature time series are allowed. In the state-space model, these missing values are treated as unobserved responses, and model estimation proceeds through Kalman filtering and smoothing.

Applying a Model Without an Exogenous Variable

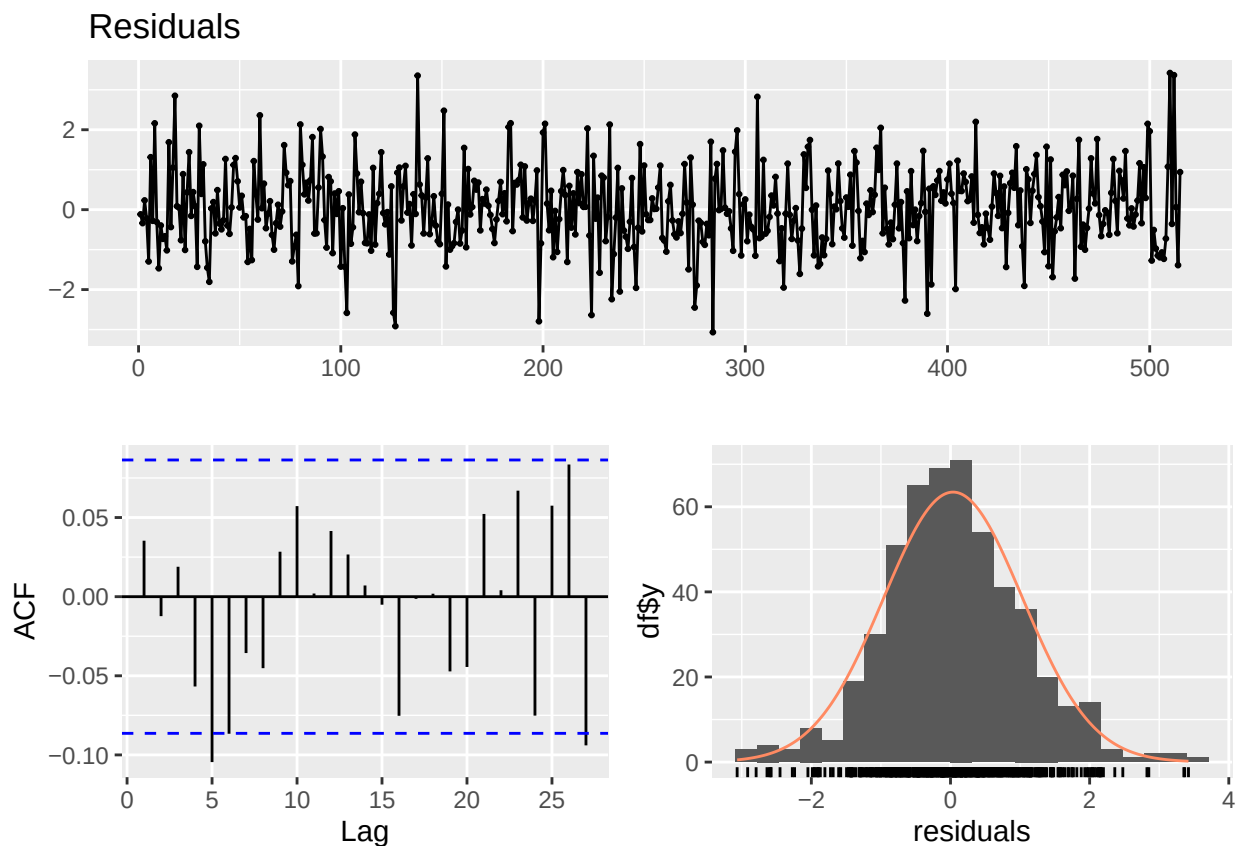
We first apply a baseline state-space model that does not include any exogenous variables. This model serves as a reference case in which temperature variation is explained solely by the latent long-term trend, seasonal cycle, and autoregressive dependence. Although the same model was already fitted in Exercise I, here we explicitly specify the dependent-variable argument and store the returned object as `res_without`.

```
res_without <- tempssm(temp_data = yamaguchi_sst_trim, ar_order = 1)
summary(res_without)
```

```
## tempssm summary
## -----
## Call:
## tempssm(temp_data = yamaguchi_sst_trim, ar_order = 1)
##
```

```
## Model fit:
## Likelihood type: marginal
## Log-likelihood : -435.46
## k : 5
## Diffuse states : 13
## Converged : TRUE
##
## Variance parameters:
## Observation (H): 2.019273e-14
## State (Q trend): 1.118445e-07
## State (Q season): 2.312771e-149
## State (Q ar): 0.3164652
##
## Components of auto-regression:
## Order of AR: 1
## Coefficient of AR1: 0.6205502
```

```
plot_tempssm_residual_diagnostics(res_without)
```



```
diag_res_without <- diagnose_residuals(res_without)
print(diag_res_without)
```

```
## # A tibble: 1 x 4
## lb_stat lb_lag lb_pvalue kurtosis
## <dbl> <dbl> <dbl> <dbl>
## 1 17.0 12 0.149 3.68
```

The residual diagnostics for this AR(1) baseline model do not indicate significant residual autocorrelation up

to lag 12. This result suggests that the default first-order autoregressive structure is a reasonable starting point for the model without the exogenous variable.

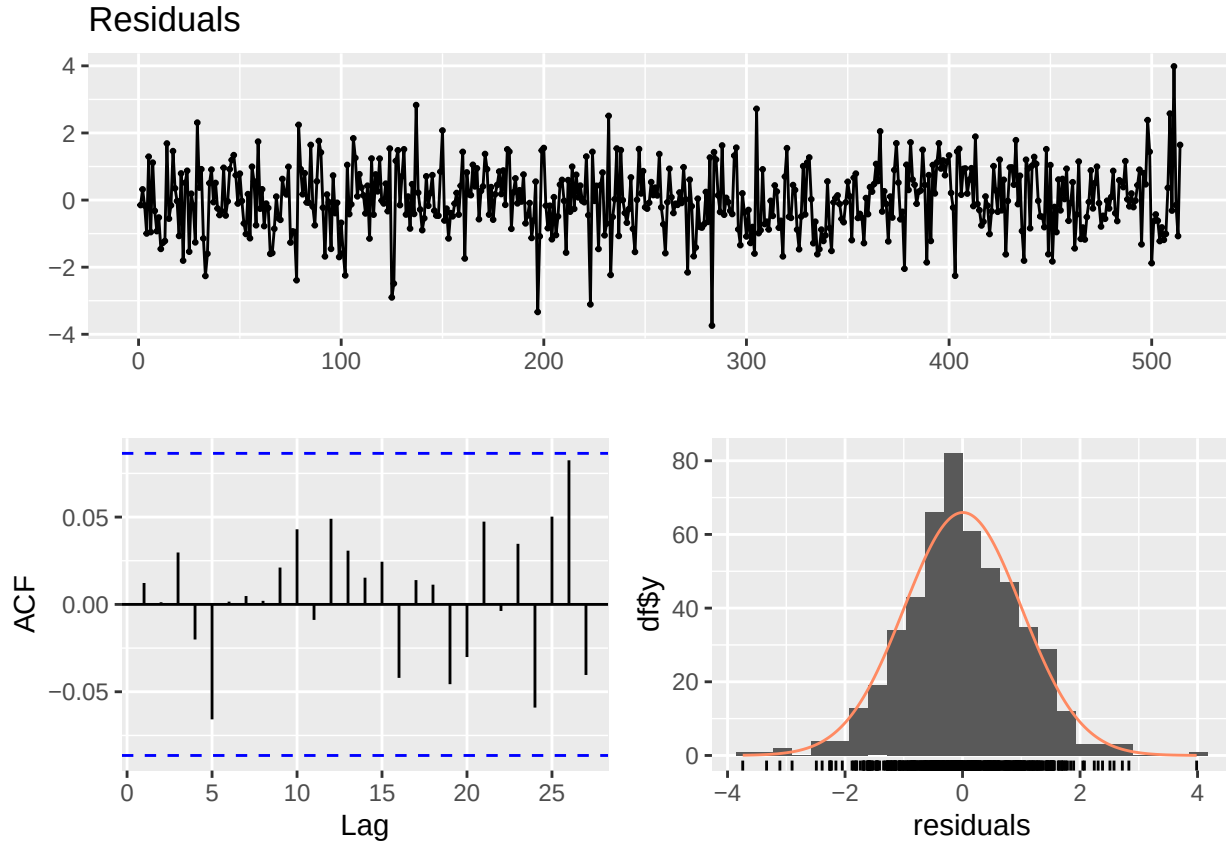
This baseline model provides a useful benchmark for evaluating the additional explanatory power of the PDO index introduced in the next section.

Applying a Model With an Exogenous Variable

We construct a model that uses the PDO index as a univariate exogenous variable. In the `tempssm()` function, the PDO data (`pdo_trim`) are supplied to the exogenous-variable argument (`exo_data`). The temperature time series (`yamaguchi_sst_trim`) is supplied to the dependent-variable argument (`temp_data`), which was implicit in the previous examples.

```
res_with <- tempssm(temp_data = yamaguchi_sst_trim,
                    exo_data = pdo_trim,
                    ar_order = 1)
summary(res_with)

## tempssm summary
## -----
## Call:
## tempssm(temp_data = yamaguchi_sst_trim, exo_data = pdo_trim,
##       ar_order = 1)
##
## Model fit:
##   Likelihood type: marginal
##   Log-likelihood : -390.99
##   k               : 6
##   Diffuse states  : 14
##   Converged       : TRUE
##
## Variance parameters:
##   Observation (H): 1.490122e-25
##   State (Q trend): 7.158619e-20
##   State (Q season): 2.316533e-82
##   State (Q ar): 0.2695508
##
## Components of auto-regression:
##   Order of AR: 1
##   Coefficient of AR1: 0.6566528
## Exogenous variable   PDO
## Estimated coefficient -0.4476073
## Lower CI  -0.5372882
## Upper CI  -0.3579264
plot_tempssm_residual_diagnostics(res_with)
```



```
diag_res_with <- diagnose_residuals(res_with)
print(diag_res_with)
```

```
## # A tibble: 1 x 4
##   lb_stat lb_lag lb_pvalue kurtosis
##   <dbl>   <dbl>   <dbl>   <dbl>
## 1     5.53     12     0.938     3.69
```

The residual diagnostics for the model with the PDO index also do not indicate significant residual autocorrelation up to lag 12. Thus, for both models considered here, the AR(1) specification provides an adequate working autoregressive structure according to these residual checks.

We then examine the model estimates. The estimated coefficient for the exogenous PDO index is negative (-0.45), and its 95% confidence interval does not include zero, ranging from -0.54 to -0.36. This indicates a statistically significant negative relationship between local SST off Yamaguchi Prefecture and PDO variability.

Specifically, after accounting for the underlying long-term trend, seasonal cycle, and autoregressive dependence, the model suggests that a one-unit increase in the PDO index is associated with an average decrease of approximately 0.45 °C in monthly SST. This result implies that positive phases of the PDO systematically contribute to cooler conditions at the study site.

Importantly, the effect of this exogenous variable is identified as an additional factor beyond the internal dynamics of the temperature time series, rather than as an artifact of improved representation of the long-term trend or temporal dependence. Compared with the baseline model without exogenous variables, the statistical significance of the PDO coefficient indicates that the PDO acts as an independent large-scale climate driver influencing local temperature variation.

For models with exogenous variables, future values of the exogenous variables are needed when calling `predict()`. If such values are available, they can be passed through `new_exo_data`. For a simple one-step-ahead visual check, `exo_strategy = "last"` carries the final observed exogenous value forward. This is a

persistence assumption for the exogenous variable, not a separate forecast model for the PDO index.

```
pred_with_last_exo <- predict(res_with, exo_strategy = "last")
pred_with_last_exo
```

```
##           Jan
## 2026 16.41488
```

Time-Series Cross-Validation (tsCV)

Time-series cross-validation (tsCV) evaluates model performance based on out-of-sample prediction errors while respecting the temporal ordering of the data. It therefore provides an additional and independent perspective on model adequacy.

In time-series cross-validation, the model is repeatedly fitted to expanding or sliding training windows, and predictive performance is evaluated on subsequent observations. Unlike random cross-validation, this procedure prevents information leakage from the future to the past and is therefore suitable for model validation with temporal data.

Here, by comparing cross-validation metrics for models with and without the exogenous PDO variable, we assess whether the statistical signal found in the PDO coefficient is also reflected in out-of-sample predictive performance.

```
# (Optional) Load packages for parallel processing.
# These are only required if you want to enable parallel execution:
# - future: controls how parallel workers are launched
# - future.apply: provides parallelized apply functions
# - progressr: optional progress bar support
library(future.apply)
library(future)
library(progressr)

if (interactive()) {
  future::plan(multisession)
} else {
  future::plan(sequential)
}

# ts cross-validation of the model without exogenous variables

## Generate a list of training and test datasets with their indices
# Procedure for constructing year-based time-series cross-validation folds:
#
# First training data: January 1982-December 2017;
# First test data: one year starting from January 2018
#
# Second training data: January 1983-December 2018;
# Second test data: one year starting from January 2019
# ...
#
# Eighth training data: January 1989-December 2024;
# Eighth test data: one year starting from January 2025
#
# These folds are automatically generated by the ts_train_test_split() function.
```

```

# Generate training and test dataset for model without PDO index
folds_without <- ts_train_test_split(
  temp_data = yamaguchi_sst_trim,
  exo_data = NULL,
  initial = 432, # 432 monthly observations from Jan 1982 to Dec 2017
  horizon = 12, # forecast 12 monthly time-series
  step = 12, # training data is moved in one-year steps
  fixed_window = TRUE,
  allow_partial = FALSE
)

# Generate training and test dataset for model with PDO index
folds_with <- ts_train_test_split(
  temp_data = yamaguchi_sst_trim,
  exo_data = pdo_trim,
  initial = 432, # 432 monthly observations from Jan 1982 to Dec 2017
  horizon = 12, # forecast 12 monthly time-series
  step = 12, # training data is moved in one-year steps
  fixed_window = TRUE,
  allow_partial = FALSE
)

# *****
# Executing time-series cross validation

#-----
# Model without the exogenous variable of PDO index

# Single processing
# cv_without_results <- ts_cv_run(folds_without, ar_order = 1, use_season = TRUE)

# Parallel processing
cv_without_results <- future.apply::future_lapply(
  folds_without,
  ts_cv_run_fold,
  ar_order = 1,
  use_season = TRUE,
  future.seed = TRUE
)

# Computing assessment indexes
metrics_without <- lapply(cv_without_results, compute_cv_metrics)

# tidy summary
cv_without_tbl <- ts_cv_collect(cv_without_results, metrics_without) %>%
  mutate(Model="Without")

#-----
# Model with the exogenous variable of PDO index

# Single processing

```

```

# cv_with_results <- ts_cv_run(folds_with, ar_order = 1, use_season = TRUE)

# Parallel processing
cv_with_results <- future.apply::future_lapply(
  folds_with,
  ts_cv_run_fold,
  ar_order = 1,
  use_season = TRUE,
  future.seed = TRUE
)

# Computing assessment indexes
metrics_with <- lapply(cv_with_results, compute_cv_metrics)

# tidy summary
cv_with_tbl <- ts_cv_collect(cv_with_results, metrics_with) %>%
  mutate(Model="With")

cv_tbl <- bind_rows(cv_without_tbl, cv_with_tbl)

cv_comparison <- compare_ts_cv(
  list(
    Without = cv_without_tbl,
    With = cv_with_tbl
  )
)

cv_comparison %>% knitr::kable()

```

model	n_folds	con- verged_n	converged_rate	mean_MAE	mean_MASE_naive	mean_MASE_seasonal
Without	8	8	1	0.6057671	0.2694420	0.7993162
With	8	8	1	0.5498521	0.2443207	0.7241733

```

plt_MAE <- ggplot(data=cv_tbl,
  aes(x=Model,y=MAE)) +
  geom_boxplot()

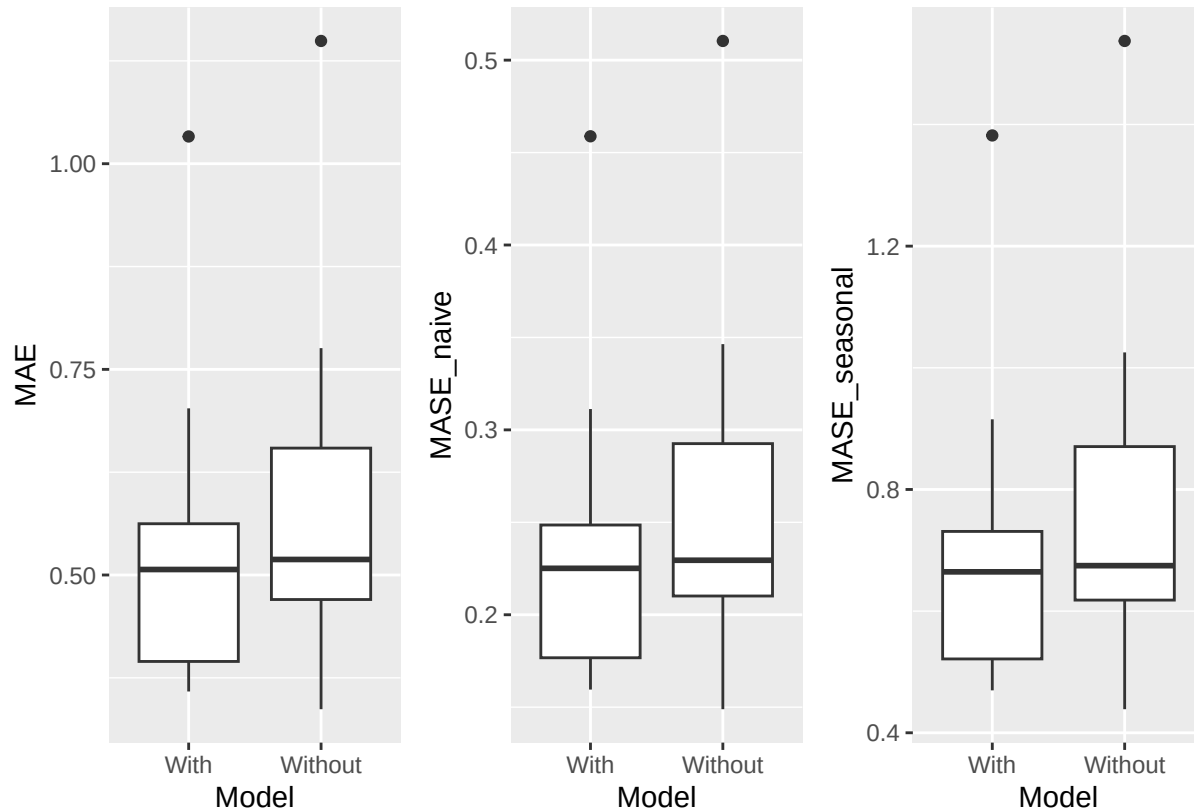
plt_MASE_naive <- ggplot(data=cv_tbl,
  aes(x=Model,y=MASE_naive)) +
  geom_boxplot()

plt_MASE_seasonal <- ggplot(data=cv_tbl,
  aes(x=Model,y=MASE_seasonal)) +
  geom_boxplot()

plt_tsCV <- plt_MAE + plt_MASE_naive + plt_MASE_seasonal + patchwork::plot_layout(nrow=1)

plot(plt_tsCV)

```



The tsCV results show that the model with the exogenous variable performs better than the model without it. The comparison table produced by `compare_ts_cv()` summarizes the number of folds, convergence rates, and mean prediction-error metrics for each model, while the boxplots show how the fold-level errors vary across validation periods. In this tutorial, the number of tsCV iterations is set to eight to reduce execution time. In actual validation, please use a sufficient number of iterations.

Taken together, the results consistently support including the PDO index as an exogenous variable in the state-space model. The PDO coefficient is statistically significant, indicating a clear effect on temperature variation. Time-series cross-validation further indicates that this effect is accompanied by better out-of-sample predictive performance.

These complementary lines of evidence provide robust and multifaceted support for adopting the exogenous-variable model.

Although this exercise used a single exogenous variable, `tempssm()` also supports models with multiple exogenous variables supplied as multivariate time-series data. Users interested in broader applications can extend the same workflow to other environmental, climatic, or ecological covariates. In doing so, model structure should be considered carefully for the target data, while checking model diagnostics and out-of-sample predictive performance.

Comparing Estimation Patterns Between Models With and Without the Exogenous Variable

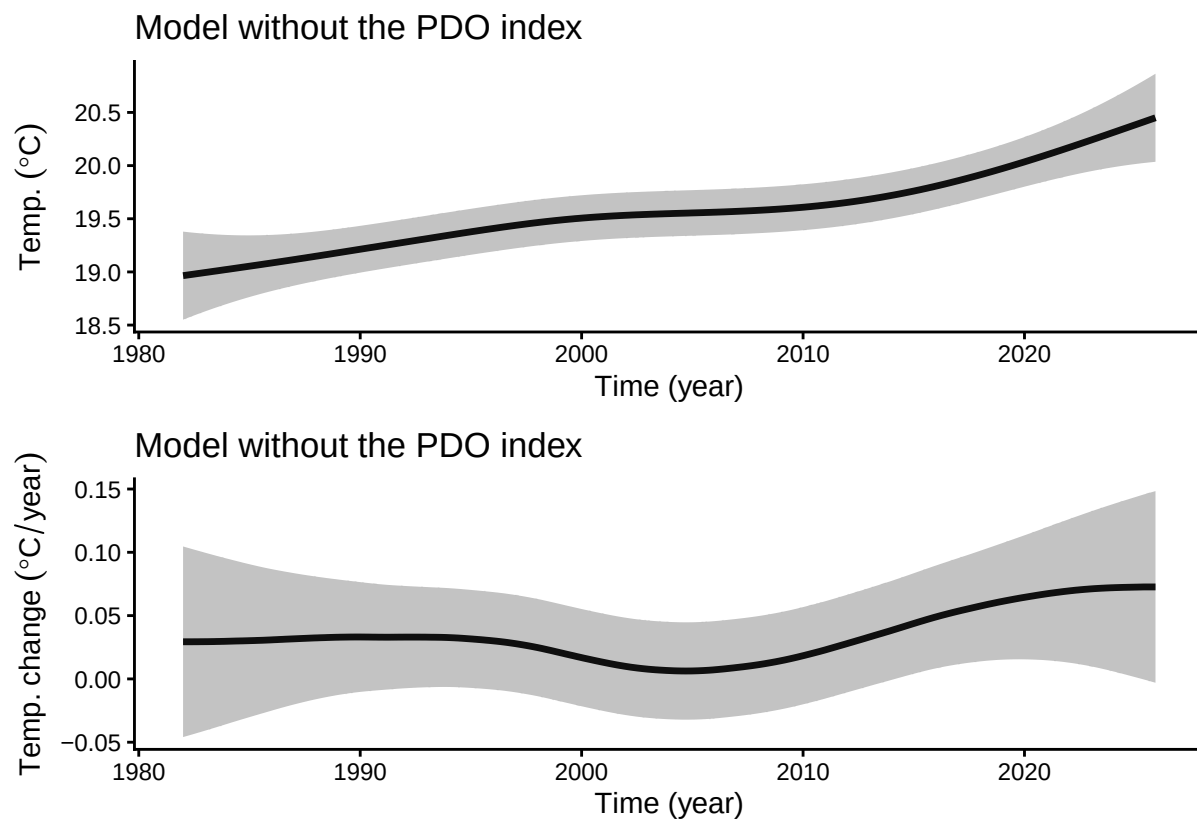
We compare how the estimated patterns of the long-term trend and its rate of change differ between the models with and without the exogenous variable.

```
plt_level_without_ts <- autoplot(res_without,
                                component=c("level")
                                ) +
  labs(title="Model without the PDO index") +
  theme_classic()
```

```
plt_drift_without_ts <- autoplot(res_without,
                                component=c("drift")
                                ) +
  labs(title="Model without the PDO index") +
  theme_classic()

plt_level_drift_without_ts <- plt_level_without_ts +
  plt_drift_without_ts +
  patchwork::plot_layout(ncol=1)

plot(plt_level_drift_without_ts)
```

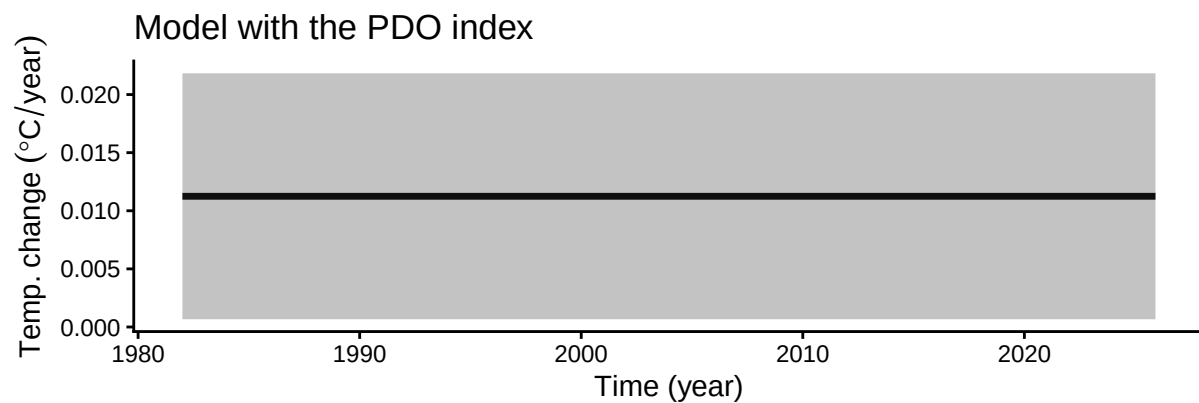
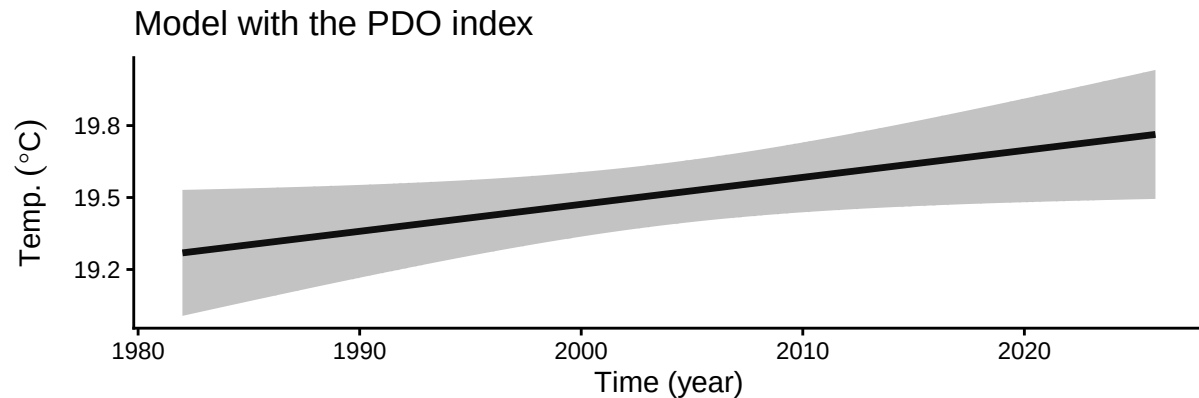


```
plt_level_with_ts <- autoplot(res_with,
                              component=c("level")
                              )+
  labs(title="Model with the PDO index") +
  theme_classic()

plt_drift_with_ts <- autoplot(res_with,
                              component=c("drift")
                              ) +
  labs(title="Model with the PDO index") +
  theme_classic()

plt_level_drift_with_ts <- plt_level_with_ts +
  plt_drift_with_ts +
  patchwork::plot_layout(ncol=1)
```

```
plot(plt_level_drift_with_ts)
```



```
# Smoothing estimate of drift component
mean_drift_year_without <- get_drift_ts(res_without) %>%
  mean()
print(mean_drift_year_without)
```

```
## [1] 0.03388663
```

```
mean_drift_year_with <- get_drift_ts(res_with) %>%
  mean()
print(mean_drift_year_with)
```

```
## [1] 0.01124586
```

The gray areas in the plots above show 95% confidence intervals.

In the model without an exogenous variable, the plots of the long-term trend and its rate of change show a wavelike pattern. In contrast, in the model with the exogenous variable, the long-term trend and its rate of change show a more stable pattern. This suggests that the wavelike pattern reflects the influence of the exogenous PDO variable.

The estimated annual rate of SST change over the observation period is 0.0339 in the model without the exogenous variable and 0.0112 in the model with the exogenous variable.

In this case, the model incorporating the PDO index is considered preferable for accurately estimating the SST trend. Thus, the SST warming rate, which was estimated somewhat higher in the model without the exogenous variable, is estimated more conservatively after introducing the model with the exogenous variable.

Using Your Own Data

Quick Example

`my_ts` is a user-supplied `ts` object containing temperature time-series data. It can be passed directly to `tempssm()`.

```
## example of ts object (not run)
## my_ts <- ts(rnorm(100), frequency = 12)
res <- tempssm(my_ts)
```

Preparing Your Own Dataset

CSV Format

Prepare a CSV file of monthly temperature time series with the following columns: - Year - Month - Temp

Example:

```
Year,Month,Temp
2010,8,13.6
2010,9,6.8
2010,10,NA
2010,11,-1.4
...
```

- Use NA for missing temperature values, and always keep the corresponding Year and Month entries.
- The CSV file must be comma-separated and UTF-8 encoded.

Converting CSV to a Time Series

An example CSV file included in this package is available in `inst/extdata`. The example dataset contains monthly air temperature observations from Mt. Akadake, Hokkaido, Japan (1,840 m). The original data are available from the Monitoring Sites 1000 Project of the Ministry of the Environment of Japan (KOZ01.zip, <https://www.biodic.go.jp/moni1000/findings/data/index.html>).

```
path <- system.file("extdata", "example_monthly_temp.csv", package = "tempssm")
akadake_temp <- tempssm::read_monthly_temp_ts(path)
head(akadake_temp)
```

```
##      Jan Feb Mar Apr May Jun Jul   Aug   Sep   Oct   Nov   Dec
## 2010              13.6   6.8   0.2  -6.8 -12.5
## 2011 -18.8
```

The `read_monthly_temp_ts()` function internally uses the base R `ts()` function to create a time-series object. If finer control is needed, you can construct the `ts` object manually.

References

The statistical modeling framework implemented in `tempssm` is based on the methodology described in Baba et al. (2024), and the accompanying supplementary materials and code repository served as the initial source for development of this package.

Baba, S., Ishii, H., and Yoshiyama, T. (2024). Estimating sea temperature trends using a linear Gaussian state-space model in Jogashima, Kanagawa, Japan. *Bulletin of the Japanese Society of Fisheries Oceanography*, 88(3), 190–199. (In Japanese with an English abstract.) https://doi.org/10.34423/jsfo.88.3_190

Baba, S. (2024). Supplementary code and test data for estimating sea temperature trends using a linear Gaussian state-space model. GitHub repository: <https://github.com/logics-of-blue/sea-temperature-trend-jogashima>

Helske, J. (2017). KFAS: Exponential Family State Space Models in R. *Journal of Statistical Software*, 78(10), 1–39. <https://doi.org/10.18637/jss.v078.i10>

Appendix: Utility Functions

The following utility functions are provided to support data preparation and exploratory analysis.

1. `read_monthly_temp_ts()`

The `read_monthly_temp_ts()` function converts monthly temperature data stored in a CSV file into an R `ts` object. By enforcing a simple and consistent data format, it is intended to make externally prepared time-series data easier to import into `tempssm`.

Example

For monthly temperature data, prepare a CSV file with a header row. By default, the column names must be `Year`, `Month`, and `Temp`, where `Temp` represents the observed temperature value.

```
Year,Month,Temp
2001,1,10.4
2001,2,8.2
2001,3,NA
2001,4,13.6
2001,5,16.1
...
```

- Use NA for missing temperature values, and always keep the corresponding Year and Month entries.
- The CSV file must be comma-separated and UTF-8 encoded.

The following example uses a sample CSV file included in the package.

```
tmp_csv <- tempfile(fileext = ".csv")
writeLines(
  c("Year,Month,Temp",
    "2001,1,10.4",
    "2001,2,8.2",
    "2001,3,NA",
    "2001,4,13.6",
    "2001,5,16.1"),
  tmp_csv
)

# Read the CSV file and convert to a monthly ts object
temp_ts <- read_monthly_temp_ts(tmp_csv)
```

2. `convert_monthly_df_to_ts()`

The `convert_monthly_df_to_ts()` function converts a data frame containing monthly temperature data into an R `ts` object. It is designed to support workflows in which temperature data are first imported or prepared as a data frame and then used for time-series analysis.

The input data frame is expected to contain at least two columns: a date column (`Date`) and a temperature column (`Temp`). For the date column (`Date`), assign an arbitrary day of the month, such as the first day, so

that the dates correspond to monthly aggregated temperature values.

Example

```
# Create a data frame of monthly temperature data
df <- data.frame(
  Date = as.Date(c(
    "2001-01-01",
    "2001-02-01",
    "2001-03-01",
    "2001-04-01",
    "2001-05-01")
  ),
  Temp = c(10.4, 8.2, NA, 13.6, 16.1)
)

# Convert to a monthly ts object
temp_ts <- convert_monthly_df_to_ts(df)
head(temp_ts)
```

```
##           Jan  Feb  Mar  Apr  May
## 2001 10.4   8.2   NA 13.6 16.1
```

3. get_jma_sst_ts()

The `get_jma_sst_ts()` function downloads daily mean sea surface temperature (SST) data for Japanese coastal waters provided by the Japan Meteorological Agency (JMA). It aggregates the daily values into monthly means and returns the resulting time series as an object of class `ts`.

Example

The following example demonstrates how to download SST data for the southern coastal waters of Ibaraki Prefecture, Japan.

The argument `sea_area_id` specifies the numeric identifier of a sea area defined by JMA. For example, 138 corresponds to the coastal waters off southern Ibaraki. A list of available sea area IDs and their corresponding regions is provided by JMA at:

https://www.data.jma.go.jp/kaiyou/data/db/kaikyo/series/engan/eg_areano.html

```
sst_138_ts <- get_jma_sst_ts(sea_area_id = 138)
head(sst_138_ts)
```

```
##           Jan      Feb      Mar      Apr      May      Jun
## 1982 15.04419 14.22500 13.63903 15.31933 17.52258 19.52300
```

4. compute_temp_anomaly()

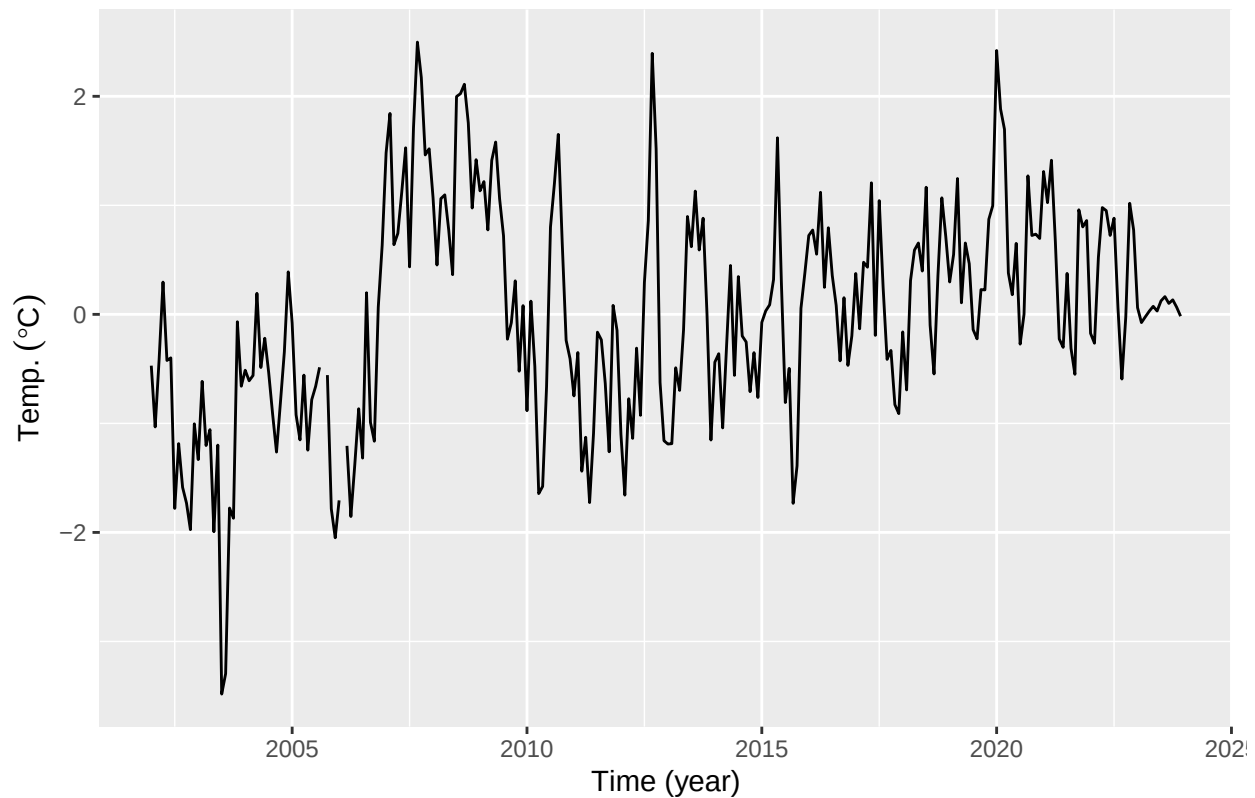
The `compute_temp_anomaly()` function computes anomalies from a time series provided as an R `ts` object. Anomalies are calculated by subtracting the corresponding long-term mean for each periodic unit, such as month, from each observation. This transformation is useful for examining interannual variability after removing periodic effects.

The reference period used to compute anomalies can be specified via a function argument. By default, the climatology is computed over the entire available time series (see `?compute_temp_anomaly`).

This transformation is useful for exploratory analysis and modeling applications that focus on departures from typical seasonal conditions.

Example

```
# Generate temperature anomalies
data(niigata_sst)
niigata_sst_anomaly <- compute_temp_anomaly(niigata_sst)
plt_niigata_sst_anomaly <- forecast::autoplot(niigata_sst_anomaly) +
  labs(y = expression(Temp.~(degree*C)),
       x = "Time (year)")
plot(plt_niigata_sst_anomaly)
```



5. compute_monthly_climatology()

The `compute_monthly_climatology()` function computes the climatological mean seasonal cycle from a `ts` object by averaging each seasonal value across years. It is primarily intended for exploratory analysis and visualization of seasonal structure in temperature time series.

Example

```
data(niigata_sst)
monthly_seasonal_cycle_niigata_sst <- compute_monthly_climatology(niigata_sst)
summary(monthly_seasonal_cycle_niigata_sst)
```

```
##      Month      Temperature
##  Min.   : 1.00   Min.    : 9.365
##  1st Qu.: 3.75   1st Qu.:11.169
##  Median : 6.50   Median :16.318
##  Mean   : 6.50   Mean    :17.036
##  3rd Qu.: 9.25   3rd Qu.:22.294
##  Max.   :12.00   Max.    :26.873
```

```
plt_monthly_seasonal_cycle_niigata_sst <- ggplot(
  data = monthly_seasonal_cycle_niigata_sst,
  aes(x = Month, y = Temperature)
) +
  geom_point(size = 2) +
  geom_line(linetype = "dashed") +
  labs(
    title = "Monthly seasonal cycle of temperature",
    x = "Month",
    y = expression(Temp.~(degree*C))
  ) +
  scale_x_continuous(
    breaks = 1:12,
    labels = sprintf("%02d", 1:12)
  ) +
  theme_classic()

plot(plt_monthly_seasonal_cycle_niigata_sst)
```

